

Spring 1990

Problem 1 (Sp90). Let $y : \mathbb{R} \rightarrow \mathbb{R}$ be a C^∞ function that satisfies the differential equation

$$y'' + y' - y = 0$$

for $x \in [0, L]$, where L is a positive real number. Suppose that $y(0) = y(L) = 0$. Prove that $y \equiv 0$ on $[0, L]$.

Problem 2 (Sp90, Fa93, Sp13, Sp18, Sp19, Sp22). 1. If A is a matrix over an algebraically closed field of characteristic 0 with $A^n = I$ for some $n > 0$ show that A is diagonalizable.

2. Show that over any field of characteristic $p > 0$ there is a matrix A , with $A^n = I$ for some $n > 0$, that is not diagonalizable.

Problem 3 (Sp90). Let $c_0, c_1, \dots, c_{n-1}$ be complex numbers. Prove that all the zeros of the polynomial

$$z^n + c_{n-1}z^{n-1} + \dots + c_1z + c_0$$

lie in the open disc with center 0 and radius

$$\sqrt{1 + |c_{n-1}|^2 + \dots + |c_1|^2 + |c_0|^2}$$

Problem 4 (Sp90). Let R be a commutative ring with identity, and R^* be its group of units. Suppose that the additive group of R is generated by $\{u^2 \mid u \in R^*\}$. Prove that R has, at most, one ideal $\mathfrak{I}$ for which $R/\mathfrak{I}$ has cardinality 3.

Problem 5 (Sp90). Suppose $x_1, x_2, x_3, \dots$ is a sequence of nonnegative real numbers satisfying

$$x_{n+1} \leq x_n + \frac{1}{n^2}$$

for all $n \geq 1$. Prove that $\lim_{n \rightarrow \infty} x_n$ exists.

Problem 6 (Sp90). Give an example of a continuous function $v : \mathbb{R} \rightarrow \mathbb{R}^3$ with the property that $v(t_1), v(t_2)$, and $v(t_3)$ form a basis for $\mathbb{R}^3$ whenever t_1, t_2 , and t_3 are distinct points of $\mathbb{R}$.

Problem 7 (Sp90, Fa92). Let a be a positive real number. Evaluate the improper integral

$$\int_0^\infty \frac{\sin x}{x(x^2 + a^2)} dx$$

Problem 8 (Sp90). Let $\mathbb{C}^*$ be the multiplicative group of nonzero complex numbers. Suppose that H is a subgroup of finite index of $\mathbb{C}^*$. Prove that $H = \mathbb{C}^*$.

Problem 9 (Sp90). Let the real valued function f on $[0, 1]$ have the following two properties:

- If $[a, b] \subset [0, 1]$, then $f([a, b])$ contains the interval with endpoints $f(a)$ and $f(b)$, i.e., f has the Intermediate Value Property.
- For each $c \in \mathbb{R}$, the set $f^{-1}(c)$ is closed.

Prove that f is continuous.

Problem 10 (Sp90, Fa93, Sp94). Show that there are at least two nonisomorphic non-abelian groups of order 24, of order 30 and order 40.

Problem 11 (Sp90, Sp22). Let the function f be analytic and bounded in the complex half-plane $\Re z > 0$. Prove that for any positive real number c , the function f is uniformly continuous in the half-plane $\Re z > c$.

Problem 12 (Sp90). Let n be a positive integer, and let $A = (a_{ij})_{i,j=1}^n$ be the $n \times n$ matrix with $a_{ii} = 2$, $a_{i,i\pm 1} = -1$, and $a_{ij} = 0$ otherwise; that is,

$$A = \begin{pmatrix} 2 & -1 & 0 & & 0 \\ -1 & 2 & -1 & & \\ 0 & -1 & 2 & & \\ & & & 0 & \\ 0 & & & -1 & 2 \end{pmatrix}$$

Prove that every eigenvalue of A is a positive real number.

Problem 13 (Sp90, Fa91). Let f be an infinitely differentiable function from $\mathbb{R}$ to $\mathbb{R}$. Suppose that, for some positive integer n ,

$$f(1) = f(0) = f'(0) = f''(0) = \cdots = f^{(n)}(0) = 0$$

Prove that $f^{(n+1)}(x) = 0$ for some x in $(0, 1)$.

Problem 14 (Sp90). Let A and B be subspaces of a finite dimensional vector space V such that $A + B = V$. Write $n = \dim V$, $a = \dim A$, and $b = \dim B$. Let S be the set of those endomorphisms f of V for which $f(A) \subset A$ and $f(B) \subset B$. Prove that S is a subspace of the set of all endomorphisms of V , and express the dimension of S in terms of n , a , and b .

Problem 15 (Sp90). Find a one-to-one conformal map of the semidisc

$$\left\{ z \in \mathbb{C} \mid \Im z > 0, \left| z - \frac{1}{2} \right| < \frac{1}{2} \right\}$$

onto the upper half-plane.

Problem 16 (Sp90). Determine the greatest common divisor of the elements of the set $\{n^{13} - n \mid n \in \mathbb{Z}\}$.

Problem 17 (Fa85, Sp90). Let f be a differentiable function on $[0, 1]$ and let

$$\sup_{0 < x < 1} |f'(x)| = M < \infty$$

Let n be a positive integer. Prove that

$$\left| \sum_{j=0}^{n-1} \frac{f(j/n)}{n} - \int_0^1 f(x) \, dx \right| \leq \frac{M}{2n}$$

Problem 18 (Sp90). Let $z_1, z_2, \dots, z_n$ be complex numbers. Prove that there exists a subset $J \subset \{1, 2, \dots, n\}$ such that

$$\left| \sum_{j \in J} z_j \right| \geq \frac{1}{4\sqrt{2}} \sum_{j=1}^n |z_j|$$